Module I – DC Circuits

Assignment Questions

1. Two resistors are connected in parallel across 200 V supply take 10 A from the mains. If the power dissipated in one resistor is 500 W. Calculate the value of the other resistor.
2. A current of 6 A flows through a 20 Ω resistor. Calculate the voltage, the power dissipated by the resistor and the energy dissipated in 600 seconds.
3. Twenty lamps each of 60 W are used for 4 hours per day in a building. Find the current drawn when all the lamps are working and also calculate the monthly electricity charge at 55 paise per unit with the supply voltage of 240 V.
4. Three resistors of 20 Ω, 30 Ω and 40 Ω are connected in series to a supply of 50 V. Calculate the equivalent resistance, current and the voltage across each resistor.
5. Find the current and voltage across the two resistors given in Figure.1.

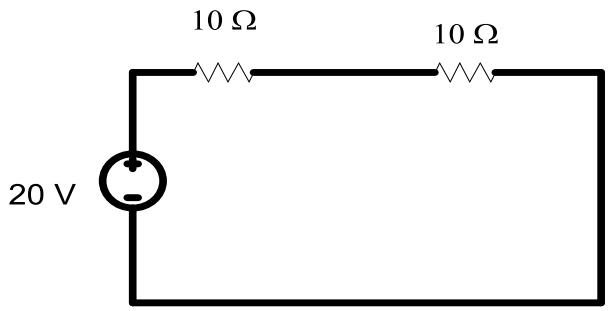

Figure.1

6. Determine the current (I) and the equivalent resistance of the circuit shown in Figure.2.

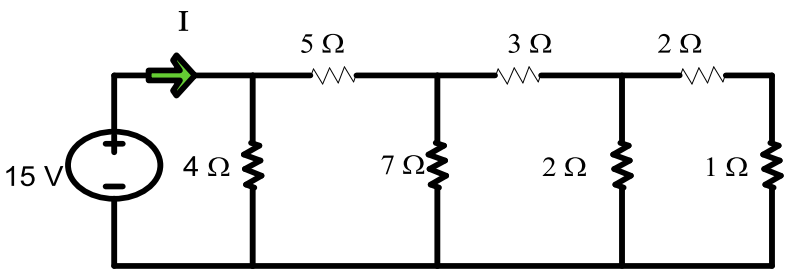

Figure.2

7. Convert a current source of 12 A with internal resistance of 3 Ω into an equivalent voltage source.

8. Convert a voltage source of 30 V with internal resistance of 3 Ω into an equivalent current source.
9. Write the mesh equations for the network shown in Figure.3 by using Inspection Method and find the power adsorbed by 6 Ω resistor.

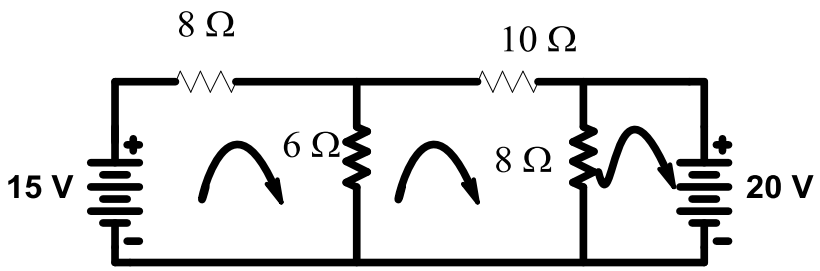

Figure.3

10. Write the mesh equations for the network shown in Figure.4 using inspection method to calculate the loop currents and also find the power absorbed by 8 Ω resistor.

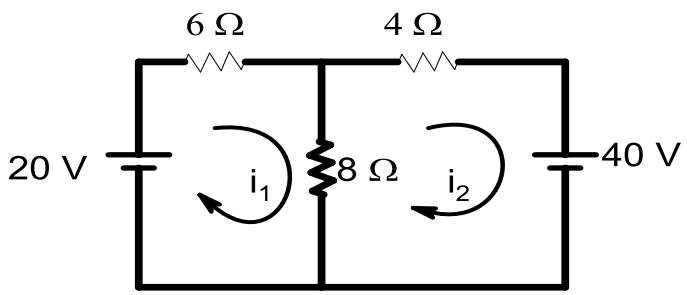

Figure.4

11. Write the mesh equations using inspection method for the network shown in Figure.5 to calculate the loop currents. Also find the power absorbed by 12 Ω resistor.

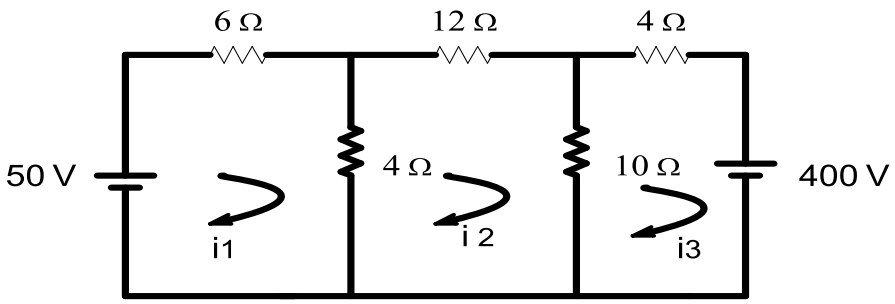

Figure.5

12. Using nodal analysis determine the current in the 20 Ω resistor is given in Figure.6.

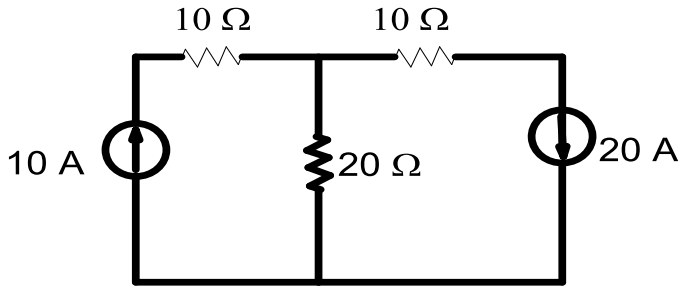

Figure.6

13. Frame the nodal equations for the network shown in Figure.7 using nodal analysis and find the potential difference between node 2 and node 4. Also find the power dissipated in 3 Ω resistor.

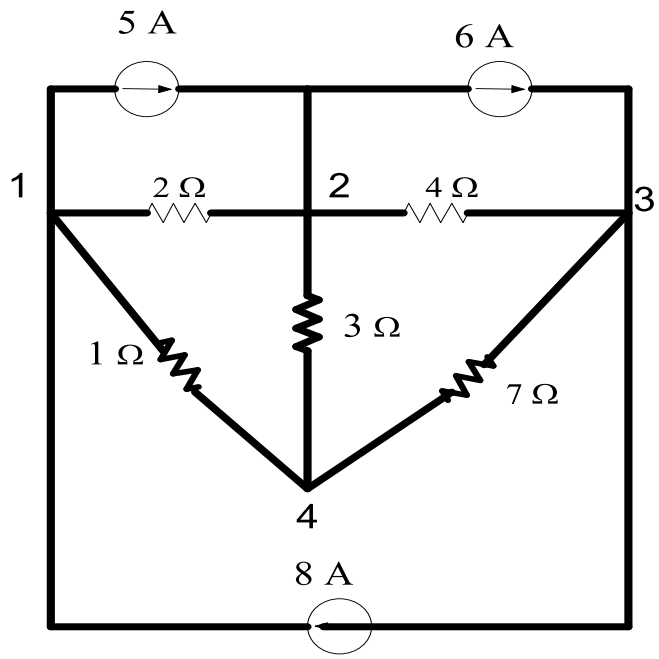

Figure.7